Josh McVey

Software Engineer · Lead SDET & Test Architect · Build & Release

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SUMMARY

Hands-on software engineer, Lead SDET, and Test Architect with 15+ years in software, spanning AI applications, robot software testing, and build and release. At Opentrons, I'm a lead software developer on the AI platform team and own release pipelines for two robot operating systems and the application suite. I work on hardware-in-the-loop testing, robot APIs, and test automation across on-robot systems, desktop applications, and cloud services. I connect product purpose, user workflows, and business goals to implementation, verification, and delivery. Interested in customer-facing work where software and physical systems meet. Lawrence, Kansas; open to remote roles with approximately 15–40% travel.

CORE SKILLS

Product & Stakeholders	User workflows, business and software objectives, requirements clarification, stakeholder alignment, acceptance criteria, engineering tradeoffs, delivery outcomes
Software & Integration	Python, TypeScript, JavaScript, Java, C#, Groovy; REST, GraphQL, OAuth/OIDC, PostgreSQL, Node, FastAPI, uv; typed robot SDKs and full-stack applications
Platform & Delivery	AWS ECS, GitHub Actions, CI/CD, Docker, Kubernetes / OpenShift, Helm, Datadog; release orchestration, OIDC npm publishing, robot diagnostics
Quality & Reliability	Lead SDET, test architecture, hardware-in-the-loop (HIL) testing, shift-left testing, testability, test framework design, API testing, pytest, Playwright/Puppeteer, Applitools, Selenium, consumer-package testing, BDD/TDD, UAT automation
AI & Technical Practice	LLM tool orchestration, Anthropic APIs, AI evaluations, retrieval-augmented generation (RAG), Amazon Bedrock, agent skills and release tooling; code review and mentoring

EXPERIENCE

Opentrons Labworks

Feb 2024 – Present

Lead Software Engineer & Build & Release Manager

Remote (HQ: Long Island City, NY)

- Own release pipelines for two robot operating systems and the application suite; built agentic tooling that enables colleagues with less technical experience to conduct releases.
- Lead software developer on the team building the Opentrons AI platform; build AI applications and tooling.
- Maintain Opentrons AI v1 on AWS ECS with Bedrock and direct model APIs; implemented bounded LLM tool loops and version-pinned retrieval context with regression tests and CI coverage checks.
- Own OAuth/OIDC authentication across web products; developed robot-interaction clients, agent skills, and asynchronous protocol analysis and simulation services.
- Added build and release responsibilities in February 2024 while continuing the software development and SDET work below.

Opentrons Labworks

Jun 2021 – Present

Senior Developer & SDET

Remote (HQ: Long Island City, NY)

- Architect, build, and operate Protocol Library (library.opentrons.com) and Opentrons Accounts, from application design through implementation.
- Built GitHub Actions/AWS delivery pipelines, OIDC npm publishing, and test frameworks that shortened robot software release cycles.
- Implemented Datadog monitoring and proactive incident detection across cloud services.
- Work on hardware-in-the-loop testing and architect pytest/Playwright and Applitools verification across robot APIs, web apps, and packed JavaScript packages; extend serial/log diagnostics and mentor engineers in testing.

TEKsystems

Aug 2019 – Jun 2021

Test Automation Architect — Continuous Testing Practice

Remote

- Practice architect for continuous testing across client engagements.
- Led pytest feature automation for a gaming-engine team of 10 (Python) and browser UAT in Azure DevOps for an energy-sector website team of 5 (JavaScript/TypeScript).

Rackspace

Nov 2016 – Aug 2019

Software Developer, Test III (Lead SDET)

Remote

- Lead SDET on a dedicated-device web portal spanning customer-facing applications, ETL pipelines, and APIs; wrote and reviewed code across teams.
- Automation and services in Python (pytest, requests, Selenium), Java (TestNG, RestAssured), TypeScript (Ava, Puppeteer), and Groovy (Grails, Spock).
- Delivery stack: Jenkins, OpenShift/Kubernetes, Helm, Docker, GraphQL, StreamSets, Elasticsearch.

Xpanxion

2009–2012; 2013–2016

Test Architect · Lead Automation Engineer · Automation Test Lead / Manager

Remote / Kearney, NE

- Hands-on QA strategy and automation consulting on long-term client engagements.
- Built frameworks for web, APIs, ETL, mobile, and desktop; BDD/TDD with Cucumber-JVM, JUnit, TestNG, NUnit, and MSTest.
- Performance testing with LoadRunner and JMeter; mentored engineers on automation and testing.

Earlier roles

2003 – 2013

Epiq Systems · Standard Beverage · Target · Independent IT

Kansas / Nebraska

- Epiq Systems (2012–2013): C# unit-test and coded-UI automation for web and desktop applications.
- Standard Beverage (2009): ASP.NET/VB.NET applications, operations dashboards, and DB2 / SQL Server database design.
- Earlier work: operations at Target; nonprofit websites, networks, databases, AV implementation, and end-user training.

EDUCATION & CERTIFICATIONS

Kansas State University

1996 – 2003

B.S., Physical Sciences

- ISTQB Certified Tester, Foundation Level (CTFL)